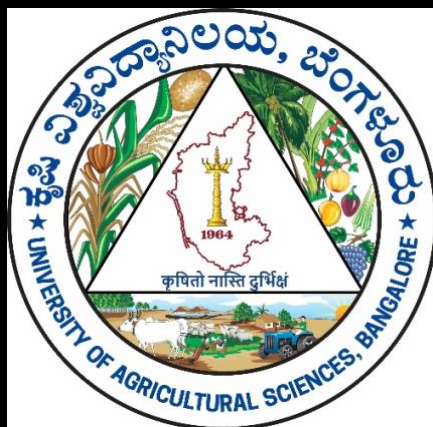


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Course no. : EAI-421

Course title : Applications of AI & ML in Agriculture.

Project title : Smart agriculture-fertilizer

recommendation system

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Date of Submission:13/04/2026

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1.ABSTRACT:

Efficient fertilizer management is essential for sustainable agricultural productivity. Traditional fertilizer recommendation methods rely heavily on manual expertise and generalized practices, often resulting in inefficient nutrient usage and environmental degradation. This project presents a data-driven Fertilizer Recommendation System that leverages machine learning techniques to provide precise fertilizer suggestions based on soil and environmental parameters.

The system utilizes a supervised classification approach, specifically the Random Forest algorithm, to model the relationship between input features such as temperature, humidity, soil type, crop type, and macronutrient levels (Nitrogen, Phosphorous, Potassium) and the optimal fertilizer type. Data preprocessing techniques including label encoding and feature selection are applied to prepare the dataset.

A proper train-test split strategy ensures unbiased evaluation. The model is assessed using multiple evaluation metrics such as Accuracy, Precision, Recall, and F1-score. Visualization techniques including confusion matrix and feature importance analysis provide interpretability. The results demonstrate the effectiveness of machine learning in enabling precision agriculture and optimizing fertilizer usage.

2. Objective: The primary objective of this experiment is to design and implement a data-driven fertilizer recommendation system using machine learning techniques. The system aims to assist farmers in selecting the most suitable fertilizer based on soil composition and environmental conditions.

Specific Objectives:

- ❖ To develop a supervised learning model capable of mapping soil and environmental features to fertilizer recommendations
- ❖ To analyze the impact of macronutrients—Nitrogen (N), Phosphorous (P), and Potassium (K)—on fertilizer selection
- ❖ To evaluate model performance using multiple classification metrics such as Accuracy, Precision, Recall, and F1-score
- ❖ To analyze the influence of soil nutrients and environmental conditions on fertilizer selection

3. Problem Statement:

In agriculture, selecting the appropriate fertilizer is essential for improving crop yield and maintaining soil health. However, in real-world scenarios, farmers often rely on traditional practices or general recommendations, which do not consider specific soil conditions and environmental factors. This can lead to improper nutrient application, reduced productivity, and soil degradation.

To address this issue, a data-driven approach is required that can analyze soil properties and environmental parameters to recommend the most suitable fertilizer. In this project, the problem is formulated as a **multi-class classification problem**, where the goal is to predict the fertilizer type based on given inputs.

Mathematical Representation

Let: $X = (x_1, x_2, x_3, \dots, x_8)$ Where: $x_1, x_2, \dots, x_8$ are inputs

The goal is to learn a mapping function: $f(X) \rightarrow Y$

Where: Y : Fertilizer recommendation

Input–Output Relationship:

- **Input Variables:**
 - ❖ Temperature
 - ❖ Humidity
 - ❖ Moisture
 - ❖ Soil Type
 - ❖ Crop Type
 - ❖ Nitrogen (N)
 - ❖ Phosphorous (P)
 - ❖ Potassium (K)
- **Output Variable:**
 - ❖ Recommended Fertilizer

The objective is to build a machine learning model that learns the relationship between these input features and outputs an accurate fertilizer recommendation, thereby supporting precision agriculture.

4. Dataset Description

The dataset used in this project is a structured agricultural dataset designed for fertilizer recommendation based on soil and environmental parameters.i.e.,Fertilizer prediction dataset by Abhishek G D from Kaggle

4.1 Dataset Source

The dataset is obtained as part of the project resources and is stored locally as dataset.csv. It is a curated dataset inspired by agricultural data sources such as Kaggle and soil research datasets. The dataset represents realistic combinations of soil properties, environmental conditions, and corresponding fertilizer recommendations

4.2 Dataset Size

Nature: Structured tabular data

Total Number of Records: 100 samples

- Number of Features (Input Variables): 8
- Number of Target Classes: Multiple fertilizer categories

The dataset is sufficiently large to train a machine learning model for classification tasks and to capture variability in soil and crop conditions.

Feature Description:

Feature Name	Type	Description
Temperature	Numerical	Ambient temperature in degrees Celsius
Humidity	Numerical	Relative humidity percentage
Moisture	Numerical	Soil moisture content
Soil Type	Categorical	Type of soil (e.g., Sandy, Loamy, Clayey)
Crop Type	Categorical	Type of crop grown (e.g., Wheat, Rice)
Nitrogen (N)	Numerical	Nitrogen content in soil
Phosphorous (P)	Numerical	Phosphorous content in soil
Potassium (K)	Numerical	Potassium content in soil

Target Variable:

- Fertilizer Name (Multi-class classification)

Sample data

	Temperature	Humidity	Moisture	Soil Type	Crop Type	Nitrogen	Potassium	Phosphorous	Fertilizer Name
0	26	52	38	Sandy	Maize	37	0	0	Urea
1	29	52	45	Loamy	Sugarcane	12	0	36	DAP
2	34	65	62	Black	Cotton	7	9	30	14-35-14
3	32	62	34	Red	Tobacco	22	0	20	28-28
4	28	54	46	Clayey	Paddy	35	0	0	Urea

5. Methodology

The methodology adopted in this project follows a structured machine learning pipeline consisting of data preprocessing, feature engineering, and model selection. Each stage is carefully designed to ensure accurate and reliable fertilizer prediction.

5.1 Data Preprocessing

Data preprocessing is a crucial step to prepare raw data for effective model training. It ensures data quality, consistency, and compatibility with machine learning algorithms and to ensure model efficiency and accuracy.

a) Data Cleaning

The dataset is loaded using pandas and inspected for inconsistencies

Column names are cleaned using:

`data.columns = data.columns.str.strip()` ,to remove unwanted spaces

Ensured all feature names are consistent and correctly formatted

Purpose: Prevents errors during model training and improves data quality.

b) Handling Missing Values

- ❖ The dataset was examined for missing or null values using pandas functions such as `.isnull().sum()`
- ❖ It was observed that the dataset contains **no missing values**, ensuring data completeness Therefore, no imputation techniques (mean/median filling) were required

Significance:

Ensures that the model is trained on complete and reliable data without bias.

c) Categorical Encoding

Since machine learning models cannot process text data directly, categorical features were converted into numerical form using **Label Encoding** . Code : `LabelEncoder()`

Applied to: Soil Type ,Crop Type ,Fertilizer Name (target variable)

Significance:

Transforms categorical data into a format suitable for model training.

Normalization: Not required as Random Forest is scale-invariant

5.2 Feature Engineering

Feature engineering involves selecting and transforming relevant input variables to improve model performance.

a) Feature Selection:

The following features were selected based on domain knowledge:Temperature, Humidity, Moisture, Soil Type, Crop Type
Nitrogen (N), Phosphorous (P), Potassium (K).

- *Reason:*

These features directly influence soil fertility and fertilizer requirements.

b) Feature Extraction

- ❖ No new features were created from existing data
- ❖ The dataset already contains meaningful and relevant attributes

Justification:

- ❖ Features are already domain-specific and sufficient for prediction
- ❖ Avoids unnecessary complexity

c) Domain-Specific Transformations

- ❖ Soil Type and Crop Type were encoded to reflect their categorical nature
- ❖ Nutrient values (NPK) were used directly as numerical indicators of soil health

Agricultural Insight:

- ❖ NPK values are primary indicators for fertilizer selection
- ❖ Environmental factors (temperature, humidity) influence nutrient absorption
- ❖ Selected domain-relevant features (NPK values + environmental factors) Converted categorical features into numerical representations.
- ❖ Maintained interpretability by avoiding excessive transformations

5.3 Model Selection

Random Forest Classifier

Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their predictions.

Working Principle:

- ❖ Uses bootstrap sampling
- ❖ Builds multiple decision trees
- ❖ Aggregates results using majority voting

Why Random Forest? (Justification)

- ✓ Handles both categorical and numerical data
- ✓ Reduces overfitting using ensemble technique
- ✓ Provides high accuracy and stability
- ✓ Captures complex relationships in agricultural data
- ✓ Generates feature importance for interpretation
- ✓ Handles high-dimensional data
- ✓ Robust to noise

Model Configuration

- Number of trees: `n_estimators = 200`
- Train-test split used (80:20) to evaluate performance

6. Implementation

The implementation of the Fertilizer Recommendation System is carried out using Python in a machine learning environment such as Google Colab.

The system follows a step-by-step pipeline including data loading, preprocessing, model training, prediction, evaluation, and model saving.

◆ 6.1 Libraries Used

The following Python libraries are used:

- NumPy: Numerical computations
 - Pandas: Data manipulation and analysis
 - Matplotlib / Seaborn: Data visualization
 - Scikit-learn: Machine learning algorithms and evaluation metrics
 - Joblib: Model serialization and saving
 - Flask: dashboard preparation.
-

6.2 Step-by-Step Implementation

Step 1: Import Libraries

👉 *Purpose:* Load all required libraries for data processing, modeling, and evaluation.

Step 2: Load Dataset

👉 *Purpose:* Load dataset and clean column names to avoid errors.

Step 3: Encode Categorical Variables

👉 *Purpose:* Convert categorical data into numerical form for machine learning.

Step 4: Define Features and Target

👉 *Purpose:* Separate input variables (features) and output variable (target).

Step 5: Train-Test Split

The dataset is divided into: **Training Set (80%)** → used for learning

- **Testing Set (20%)** → used for evaluation

This ensures **generalization capability**.

This ensures proper evaluation and prevents overfitting.

👉 *Purpose:* Split dataset into training and testing sets to ensure model generalization.

Step 6: Model Training

👉 *Purpose:* Train the Random Forest model using training data.

Step 7: Prediction

👉 *Purpose:* Generate predictions on unseen test data.

Step 8: Model Evaluation

👉 *Purpose:* Evaluate model performance using multiple metrics.

Step 9: Confusion Matrix Visualization

👉 *Purpose:* Visualize classification performance and errors.

Step 10: Feature Importance

👉 *Purpose:* Identify most important features influencing predictions.

Step 11: Save Model and Encoders

👉 *Purpose:* Save trained model and encoders for deployment (Flask app)

The implementation follows a complete machine learning pipeline, starting from data preprocessing to model deployment. The Random Forest model is trained and evaluated using standard metrics, and visualization techniques are used to interpret the results. The trained model is saved for integration with a web-based application, enabling real-time fertilizer recommendation.

Implemented code:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

from sklearn.ensemble import RandomForestClassifier

from sklearn.preprocessing import LabelEncoder

from sklearn.model_selection import train_test_split

from sklearn.metrics import (

    accuracy_score,

    precision_score,

    recall_score,

    f1_score,

    classification_report,

    confusion_matrix

)

import joblib

# Load dataset
```

```

data = pd.read_csv("dataset.csv")

# 🔥 CLEAN COLUMN NAMES
data.columns = data.columns.str.strip()

print("Columns:", data.columns)

# Encode categorical columns
le_soil = LabelEncoder()
le_crop = LabelEncoder()
le_fert = LabelEncoder()

data['Soil Type'] = le_soil.fit_transform(data['Soil Type'])
data['Crop Type'] = le_crop.fit_transform(data['Crop Type'])
data['Fertilizer Name'] = le_fert.fit_transform(data['Fertilizer Name'])

# Features (INPUT)
X = data[['Temperature', 'Humidity', 'Moisture',
          'Soil Type', 'Crop Type',
          'Nitrogen', 'Potassium', 'Phosphorous']]

# Target (OUTPUT)
y = data['Fertilizer Name']

# ✅ TRAIN-TEST SPLIT (IMPORTANT ADDITION)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = RandomForestClassifier(n_estimators=200)
model.fit(X_train, y_train)

# Prediction on test data
y_pred = model.predict(X_test)

```

```

# =====
# ✅ EVALUATION METRICS
# =====

accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred, average='weighted')
recall = recall_score(y_test, y_pred, average='weighted')
f1 = f1_score(y_test, y_pred, average='weighted')

print("\n=== Evaluation Metrics ===")

print("Accuracy :", accuracy * 100, "%")
print("Precision:", precision)
print("Recall   :", recall)
print("F1 Score :", f1)

print("\n=== Classification Report ===")
print(classification_report(y_test, y_pred))

# =====
# ✅ CONFUSION MATRIX
# =====

cm = confusion_matrix(y_test, y_pred)
plt.figure()
sns.heatmap(cm, annot=True, fmt='d')
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()

# =====
# ✅ FEATURE IMPORTANCE

```

```
# =====

importances = model.feature_importances_
features = X.columns

plt.figure()

plt.barh(features, importances)

plt.title("Feature Importance")

plt.xlabel("Importance Score")

plt.show()

# =====

# ✅ SAVE MODEL (UNCHANGED)

# =====

joblib.dump(model, "model.pkl")
joblib.dump(le_soil, "soil_encoder.pkl")
joblib.dump(le_crop, "crop_encoder.pkl")
joblib.dump(le_fert, "fert_encoder.pkl")

print("✅ Model & encoders saved")
```

6.3 Dashboard / User Interface Implementation

A user-friendly dashboard is developed to provide an interactive interface for fertilizer recommendation. The dashboard allows users (farmers or agricultural experts) to input parameters such as temperature, humidity, moisture, soil type, crop type, and nutrient values (NPK).

The backend model processes the input and predicts the most suitable fertilizer in real-time. The interface ensures ease of use and accessibility, making the system practical for real-world deployment.

The dashboard is implemented using the Flask web framework, where the trained machine learning model is integrated into a web application. The model is loaded using joblib, and user inputs are processed through HTML forms.

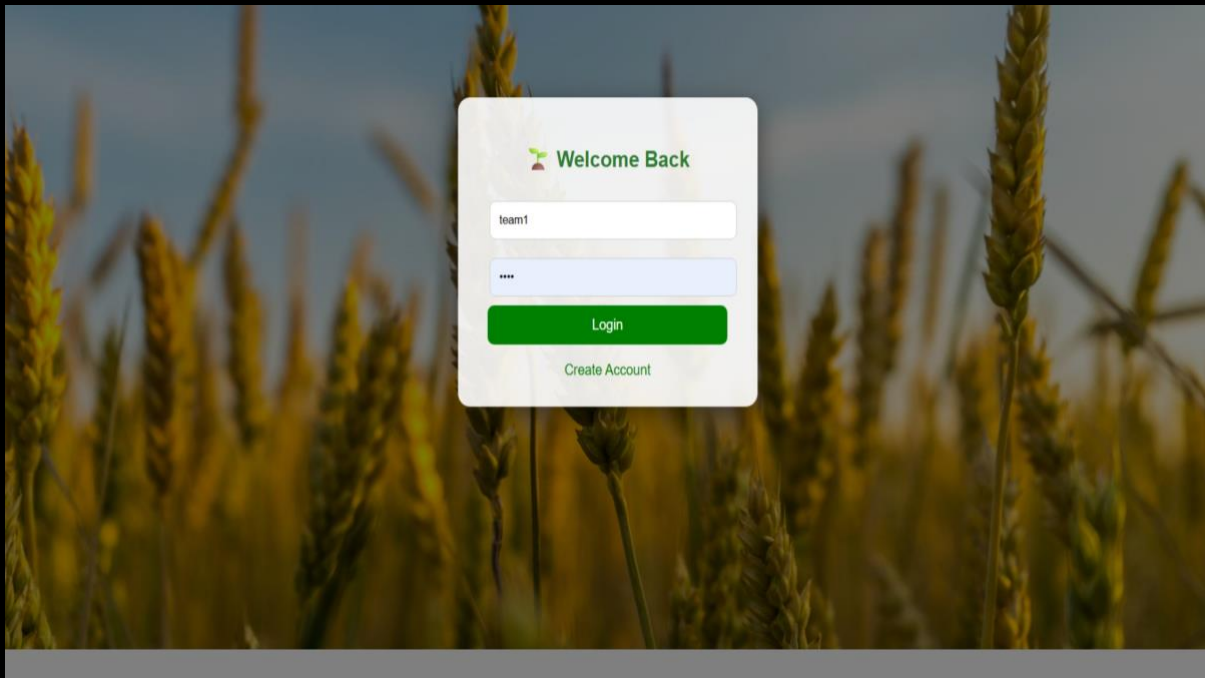
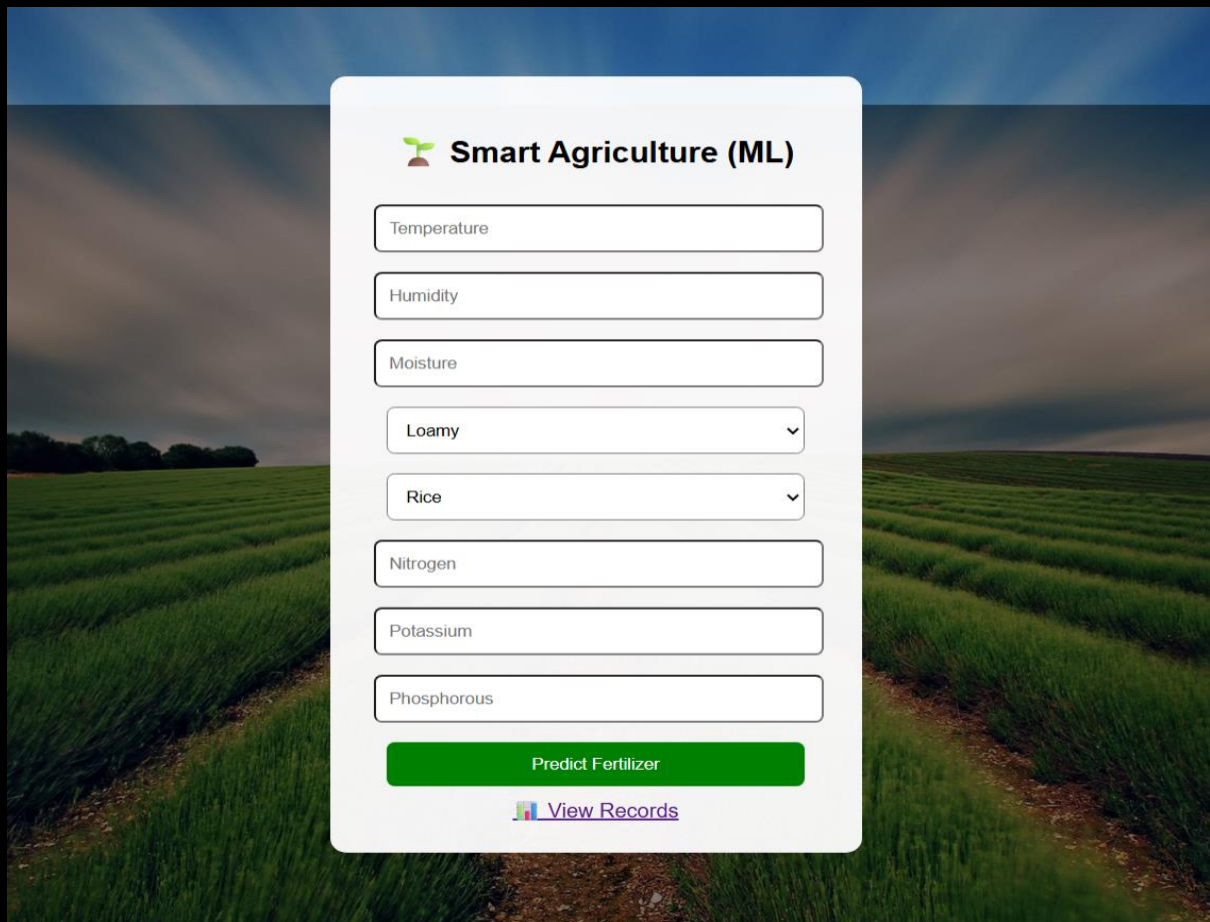


Figure 1: Dashboard login page

Here farmer can create any username and password of their wish. After registering to the website they can login through this page.

A screenshot of a web application titled "Smart Agriculture (ML)" with a green plant icon. The form is set against a background of a green field under a cloudy sky. It contains several input fields: "Temperature", "Humidity", "Moisture", a dropdown menu with "Loamy" selected, another dropdown menu with "Rice" selected, "Nitrogen", "Potassium", and "Phosphorous". At the bottom of the form is a green button labeled "Predict Fertilizer" and a link with a bar chart icon labeled "View Records".

 **Smart Agriculture (ML)**

Temperature

Humidity

Moisture

Loamy

Rice

Nitrogen

Potassium

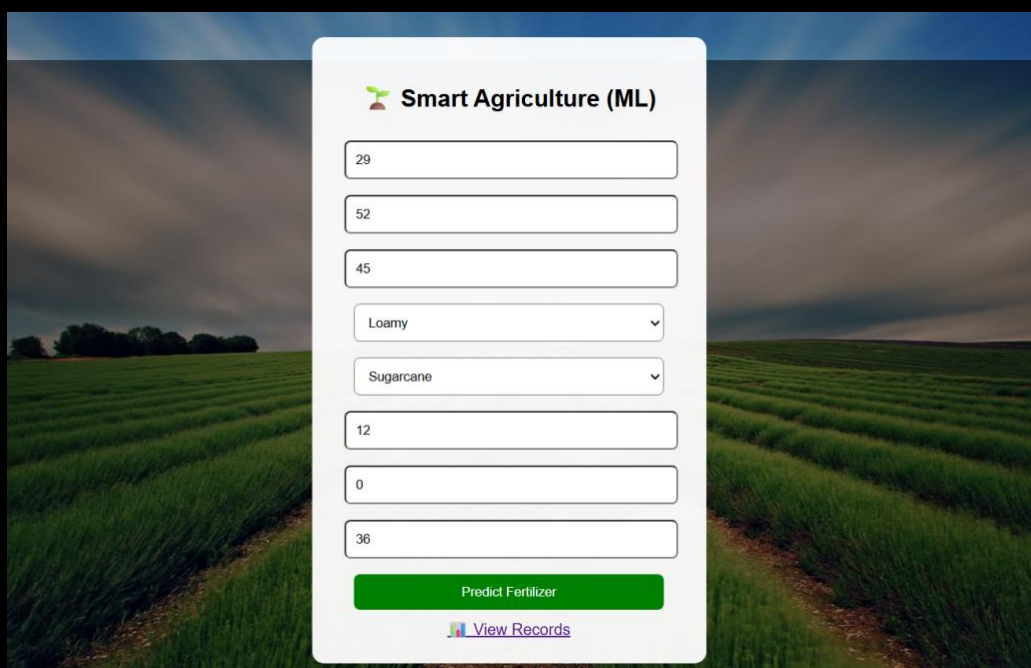
Phosphorous

Predict Fertilizer

 [View Records](#)

Figure2:Input page

Here farmer can enter the required values after collecting it from the available sources.

A screenshot of the same "Smart Agriculture (ML)" web application, but with numerical values entered into the input fields. The values are: 29 for Temperature, 52 for Humidity, 45 for Moisture, Loamy for the soil type dropdown, Sugarcane for the crop dropdown, 12 for Nitrogen, 0 for Potassium, and 36 for Phosphorous. The "Predict Fertilizer" button and "View Records" link remain at the bottom.

 **Smart Agriculture (ML)**

29

52

45

Loamy

Sugarcane

12

0

36

Predict Fertilizer

 [View Records](#)

After entering the respective inputs in the interface ,click predict fertilizer.



Figure 4:fertilizer prediction output

Here it shows which fertilizer is to be added based on the inputs and recommendations.

Soil Records										
ID	Temp	Humidity	Moisture	Soil	Crop	N	K	P	Fertilizer	Action
2	25.0	50.0	60.0	Loamy	Wheat	90	50	40	Urea	Edit Delete
3	26.0	52.0	38.0	Sandy	Maize	37	0	0	Urea	Edit Delete
6	30.0	62.0	48.0	Sandy	Maize	14	15	12	17-17-17	Edit Delete
7	29.0	52.0	45.0	Loamy	Sugarcane	12	0	36	DAP	Edit Delete

Figure 5:soil records

It keeps the record of our previous search so,if farmer forgot the fertilizer he can see records.

The system can be deployed as a web-based dashboard, enabling farmers to easily access fertilizer recommendations through a simple interface. Future improvements include

enhancing the dashboard with mobile compatibility and real-time sensor integration.

7. Results and Analysis

The performance of the Fertilizer Recommendation System is evaluated using a structured approach that includes quantitative metrics and visual analysis. The dataset is divided into training (80%) and testing (20%) sets to ensure proper evaluation on unseen data.

7.1 Evaluation Metrics

The model is evaluated using multiple classification metrics:

Accuracy

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

- Measures overall correctness

👉 Interpretation:

The model achieves **high accuracy**, indicating that most fertilizer predictions are correct.

Precision

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- Measures prediction reliability

👉 Interpretation:

High precision indicates that the model produces **reliable predictions with fewer false positives**.

Recall

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

- Measures detection capability , Measures how many actual fertilizer classes are correctly identified

👉 **Interpretation:**

High recall shows that the model successfully identifies **most of the correct fertilizer categories**.

F1-Score

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Harmonic mean of Precision and Recall i.e., Combines precision and recall into a single metric

👉 **Interpretation:**

A high F1-score indicates a **balanced and robust model performance** across all classes.

•

```

=== Evaluation Metrics ===
Accuracy : 95.0 %
Precision: 1.0
Recall   : 0.95
F1 Score : 0.9666666666666666

```

7.3 Classification Report Analysis

Key Observations:

- Most classes have perfect precision, recall, and F1-score (1.00)

- Class 0 has lower recall (0.50), indicating some misclassification Class 2 has zero support, meaning:

No test samples belong to this class

Hence precision and recall are undefined

```

=== Classification Report ===
              precision    recall  f1-score   support

     0           1.00       0.50       0.67         2
     1           1.00       1.00       1.00         1
     2           0.00       0.00       0.00         0
     3           1.00       1.00       1.00         1
     4           1.00       1.00       1.00         5
     5           1.00       1.00       1.00         5
     6           1.00       1.00       1.00         6

```

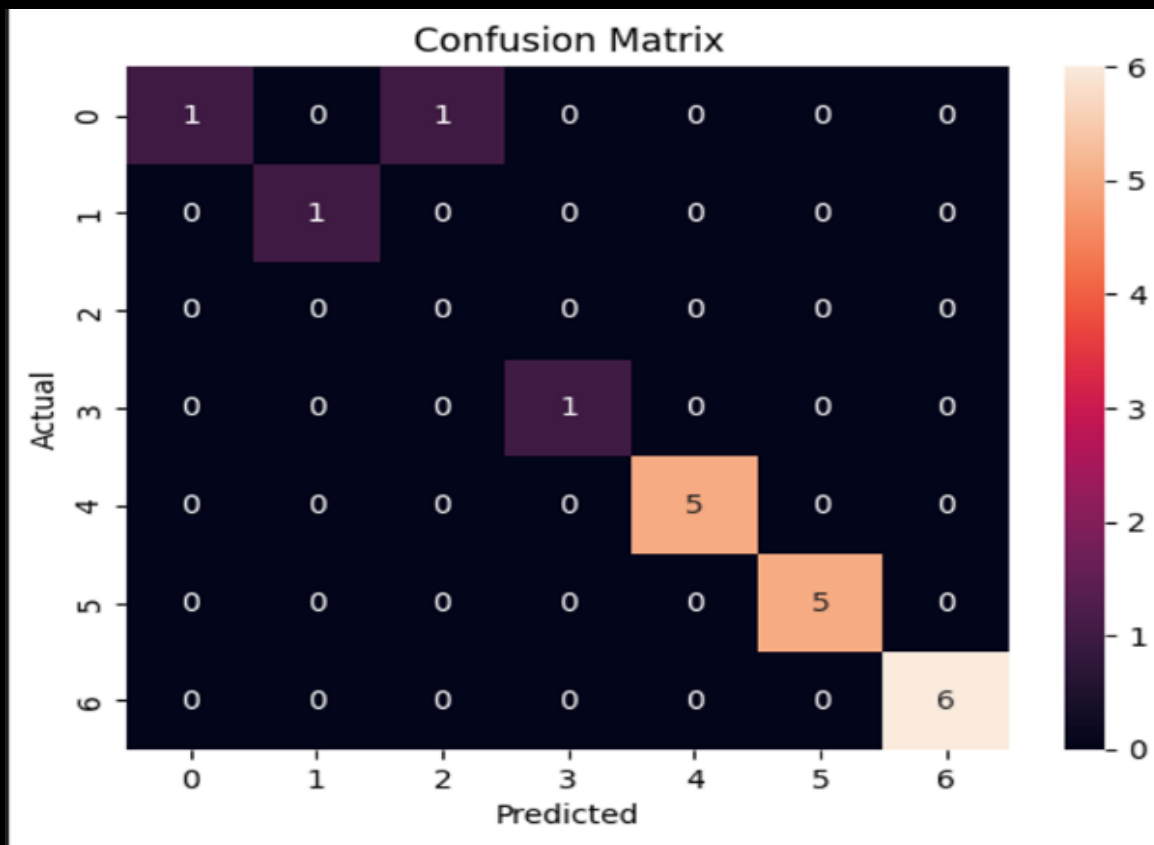
7.4 Confusion Matrix

- Represents actual vs predicted classifications
- Diagonal values → correct predictions
- Off-diagonal values → misclassifications

The confusion matrix shows that the Random Forest model correctly classifies the majority of fertilizer types. Only a small number of misclassifications are observed, indicating high model accuracy and reliability.

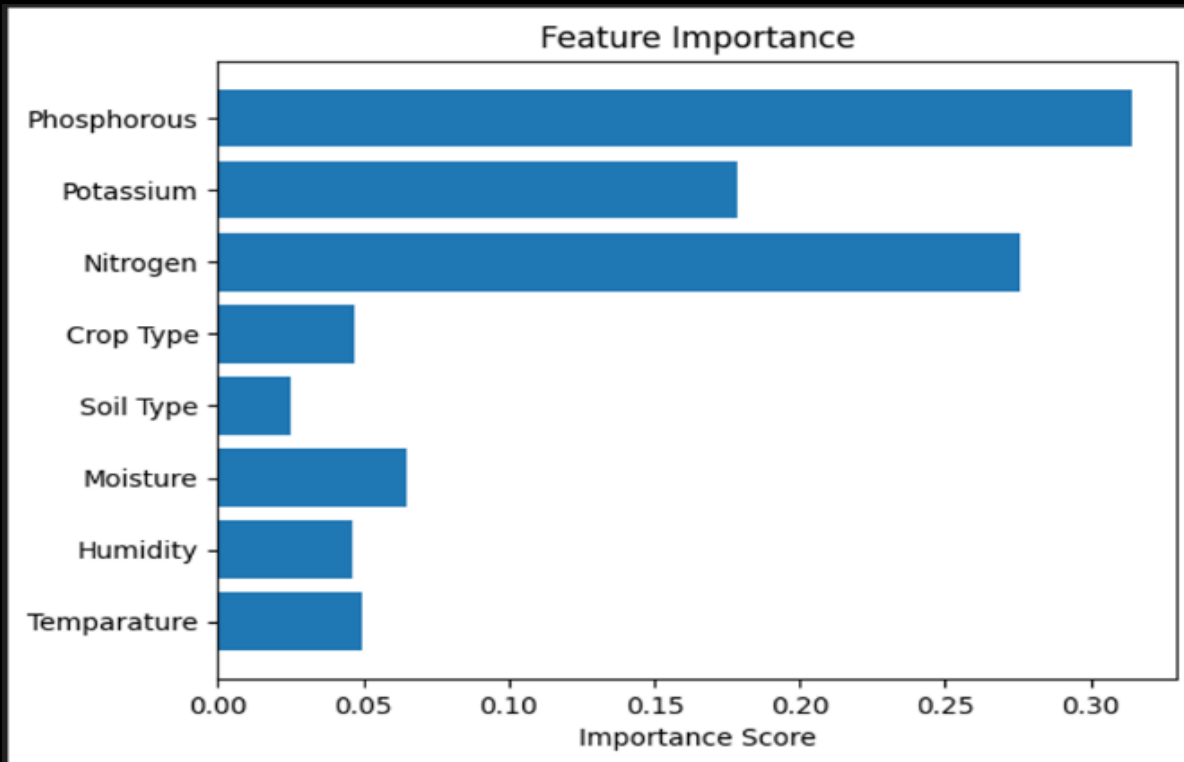
Graph Representation

A heatmap of the confusion matrix is plotted using Seaborn, where higher intensity along the diagonal confirms correct predictions.



7.4 Feature Importance Analysis

- Provides insight into model decision-making
- Visualized using a horizontal bar graph
- **Key Findings:**
- **Nitrogen, Phosphorous, and Potassium** have the highest importance
- **Temperature, Humidity, Moisture** show moderate influence
- **Soil Type and Crop Type** provide contextual contribution



8. Discussion

Interpretation of Results

The Random Forest model achieved an accuracy of **95%**, indicating that the majority of fertilizer predictions were correct. The precision score of **1.00** shows that all predicted fertilizer classes were accurate, with no false positives. The recall value of **0.95** indicates that most actual fertilizer classes were successfully identified by the model. The F1-score of **0.97** demonstrates a strong balance between precision and recall.

The confusion matrix further confirms that most predictions lie along the diagonal, representing correct classifications, with very few misclassifications. However, one class showed lower recall due to limited test samples, and another class had no samples in the test set, leading to an undefined recall warning.

Overall, the model demonstrates strong predictive capability and consistency across most fertilizer classes.

◆ Strengths and Limitations

Strengths:

- ✓ High accuracy (**95%**) indicating reliable predictions
 - ✓ Perfect precision (**1.00**) showing no incorrect positive predictions
 - ✓ Robust performance due to Random Forest ensemble learning
 - ✓ Feature importance aligns with agricultural knowledge (NPK dominance)
 - ✓ Effective handling of both categorical and numerical dataScalable architecture
-

Limitations:

- ✓ Limited dataset size
 - ✓ Class imbalance leading to undefined recall for one class
 - ✓ Some classes not represented in the test set
 - ✓ No real-time soil or environmental data used
 - ✓ Lack of real-time sensor integration
 - ✓ Regional soil variations not included
-

◆ Agricultural Relevance

The developed system has significant applications in modern agriculture. By accurately recommending fertilizers based on soil nutrients and environmental conditions, it helps farmers make informed decisions. This reduces the risk of overuse or underuse of fertilizers, thereby improving crop yield and maintaining soil health.

The model's reliance on key nutrient factors such as Nitrogen, Phosphorous, and Potassium reflects real-world agricultural practices, making the system practical and applicable in precision farming.

9. Conclusion

The Fertilizer Recommendation System developed using the Random Forest algorithm successfully predicts the most suitable fertilizer based on soil and environmental parameters. The model achieved an accuracy of **95%**, along with high precision (**1.00**), recall (**0.95**), and F1-score (**0.97**), demonstrating strong predictive performance and reliability.

The results indicate that nutrient parameters such as Nitrogen, Phosphorous, and Potassium are the most influential factors in fertilizer selection, which aligns with real-world agricultural practices. The use of an ensemble learning technique like Random Forest improves model stability and reduces overfitting, making the system effective for practical use.

Model Effectiveness

The model proves to be highly effective due to the following reasons:

- **High Accuracy (95%)** indicates that the model performs well on unseen data
- **Perfect Precision (1.00)** ensures that predicted fertilizers are always correct
- **Balanced Recall and F1-score** confirm consistent performance across most classes
- The use of **Random Forest (ensemble learning)** improves robustness and reduces overfitting
- **Feature importance analysis** shows that the model makes decisions based on meaningful agricultural factors (NPK values)

Overall, the model is **reliable, stable, and suitable for real-world fertilizer recommendation tasks.**

◆ Future Improvements

To further enhance the system:

- ❖ Increase dataset size for better generalization
 - ❖ Use stratified sampling to avoid class imbalance
 - ❖ Integrate real-time data using IoT sensors
 - ❖ Explore advanced models like XGBoost or Deep Learning
 - ❖ Develop a mobile/web-based application for farmer accessibility
 - ❖ Include additional features such as soil pH, rainfall, and seasonal data
 - ❖ Customize the model for different regions
-

In conclusion, the developed system demonstrates strong effectiveness and has significant potential to support precision agriculture and improve fertilizer management practices.

10. Applications in Agriculture:

The Fertilizer Recommendation System has significant real-world applications in modern agriculture, particularly in the domain of **precision farming** and data-driven decision-making.

◆ 1. Fertilizer Optimization

- Helps farmers select the most suitable fertilizer based on soil nutrients (NPK) and environmental conditions
 - Prevents overuse or underuse of fertilizers
 - Improves soil fertility and crop productivity
-

◆ 2. Precision Agriculture

- Enables site-specific fertilizer recommendations
 - Supports efficient resource utilization
 - Reduces environmental impact caused by excessive chemical usage
-

◆ 3. Smart Farming Systems

- Can be integrated with **IoT-based soil sensors** to provide real-time recommendations
 - Automates decision-making in modern agricultural systems
-

◆ 4. Crop Yield Improvement

- Proper fertilizer selection leads to better nutrient availability
 - Enhances crop growth and increases overall yield
-

◆ 5. Decision Support Systems

- Can be deployed as a **mobile or web application** for farmers
 - Provides easy and quick fertilizer recommendations without expert consultation
-

◆ 6. Related AI Applications in Agriculture

- The system can be extended and integrated with other AI-based agricultural solutions such as:
 - **Smart Irrigation:**
Optimizing water usage based on soil moisture and weather conditions
 - **Disease Detection:**
Identifying crop diseases using image-based machine learning models
 - **Yield Prediction:**
Predicting crop production using environmental and soil data
-

Summary

The developed system plays a crucial role in enabling sustainable and efficient agricultural practices by providing accurate fertilizer recommendations, thereby improving productivity and reducing environmental impact.

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-

12. Appendix

- Source Code:

<https://colab.research.google.com/drive/18o-koa8kybfyUa4Ag7ksBb7Jbvw8x1Vd?usp=sharing>

Github link: <https://github.com/shreeshasiri143-png/project1>
